

# Interpretable Attention for Quantifying Player Contribution in Football Event Sequences

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**Abstract.** Quantifying the contribution of football players from event data remains challenging, as valuable actions are often context-dependent and distributed across sequences of play rather than captured by isolated events. This paper proposes an attention-based Transformer framework for estimating player contribution in shot-ending possession sequences. The model predicts the expected-goals value of the final shot and uses attention weights assigned to preceding events to derive event- and player-level contribution scores.

The empirical evaluation is conducted using event data from the 2025/2026 Polish Ekstraklasa season. The dataset comprises 306 matches, 8,441 shot-ending sequences, and actions involving 517 players. This setting enables the analysis of player contributions across the analyzed competition and season.

We evaluate the proposed framework through predictive performance, ablation experiments, and player-level ranking stability. The results show that event context, especially action-outcome information, plays an important role in predicting shot value, whereas player identity and spatio-temporal information affect the structure and stability of attention-based contribution estimates. The study demonstrates the feasibility of using attention mechanisms as an interpretable tool for football event-sequence analysis, while also highlighting the need for caution when interpreting attention weights as direct causal explanations. The proposed approach offers a practical basis for decision-support tools in sports analytics, particularly for identifying players who contribute to valuable attacking sequences beyond final shots and assists.

**Keywords:** Football analytics · Expected goals · Attention mechanisms · Explainable AI · Player contribution · Transformer models

## 1 Introduction

Football analytics increasingly uses event data, tracking data, and machine learning models to support tactical analysis, scouting, player evaluation, and decision-making [5,26,24]. Metrics such as goals, assists, shots, and expected goals (xG) are now standard tools for evaluating attacking performance [16,22,1]. However, they primarily summarize final outcomes or final actions. They do not fully answer a question central to analysts and coaches: which players and which earlier actions contributed to creating a shooting opportunity?

Action-valuation approaches such as VAEP, expected threat, and on-ball value address this limitation by assigning value to passes, carries, and other on-ball actions [3,18,19]. These methods have become influential because they move beyond goals and shots. Nevertheless, they usually define contribution through a hand-designed state-value framework or through action-level marginal changes. In contrast, the present paper investigates whether the internal attention structure of a sequence model can be used as an interpretable, stress-testable signal of contribution within complete shot-ending event sequences.

The use of attention for explanation is controversial. Attention weights can be useful diagnostic signals, but they should not be equated with causal explanations [7,25]. This is particularly important in football analytics, where model outputs may influence scouting, tactical assessment, or player evaluation. A high attention weight assigned to a player’s action should therefore be interpreted as model reliance within a specific representation, not as proof that the player caused the shot quality to increase.

This paper is motivated by these issues and is deliberately framed as an exploratory case study rather than as a fully validated player-rating system. We train a Transformer-style sequence model on shot-ending event sequences and use the source-provided xG value of the final shot as a representation-learning target. The learned attention maps are then aggregated into player-level contribution scores. Importantly, because the final shot token is included in the sequence and may contain information strongly related to the xG target, we do not claim to build an independent pre-shot xG predictor. Instead, we study a post-hoc explanation setting: given an observed shot-ending sequence, how does the model distribute attention across its events and players, and how stable are the resulting rankings under feature ablations?

The main contributions are:

- a transparent formulation of attention-based player contribution for shot-ending football event sequences;
- an explicit discussion of the target–feature relationship and the risk of information leakage when source-provided xG is used as a training target;
- an empirical audit of attention-derived contribution scores using validation error, attention entropy, Spearman rank stability, and Jaccard@10 stability;
- an ablation study over player identity, spatial context, temporal dynamics, and outcome information;
- visual outputs that show how attention-derived scores can support, but not replace, expert football analysis.

## 2 Related Work

### 2.1 Event-Based Football Analytics and Action Valuation

Event-based football analytics aims to describe and evaluate match actions using structured data. Each event typically includes information on the action type, the player, the team, the pitch location, the time, and the outcome. Such data have

been widely used to develop metrics for evaluating passes, shots, possessions, player involvement, and the tactical structure of match play [8,5].

Expected goals (xG) is one of the most widely used metrics in football analytics. It estimates the probability that a shot will result in a goal, usually based on contextual information such as shot location, angle, body part, and type of assist [16,22]. Recent work has also emphasized the need for accurate and explainable xG models, especially when used in practical decision-support settings [1]. While xG is highly useful as a shot-quality metric, it primarily focuses on the final action in an attacking move. It does not fully explain how previous events and players contributed to the creation of the opportunity.

Several frameworks have been proposed to assign value to actions beyond shots. VAEP estimates how individual actions change the probability of scoring and conceding in the subsequent game state [3]. Expected threat models value ball movement by estimating how actions change the likelihood of eventually scoring from different pitch zones [18]. On-ball value is another influential action-valuation approach used in professional football analytics [19]. Comparative studies of xT and VAEP show that different action-value metrics capture different aspects of on-ball contribution [17]. These methods provide important tools for valuing passes, carries, and other events. However, many of them evaluate events individually or locally, rather than explicitly modeling the entire sequence leading to a shot.

Player-level evaluation is also closely related to the present work. PlayeRank introduced a data-driven framework for role-sensitive player performance ranking [12], while subsequent studies have explored technical profiles and league-specific differences in player evaluation [13,14]. In contrast to these approaches, our framework derives contribution scores from attention distributions learned directly over shot-ending event sequences.

## 2.2 Sequential and Pattern-Based Modeling in Football

Football is inherently sequential. The value of an action depends on what happened before, where the ball is located, which players are involved, and how the defensive structure reacts. Sequence-based analysis has therefore become an important direction in football analytics. Previous studies have used clustering and pattern-recognition techniques to identify typical buildup structures and tactical patterns [6,10]. SoccerMix represents soccer actions using mixture models and highlights the importance of structured representations of event sequences [4]. Deep learning has also been applied to value defensive and offensive actions from event data, including sequence definitions that explicitly model the context preceding a target action [9,11].

Recent surveys show that deep learning in sports increasingly encompasses perception, comprehension, and decision support tasks [26]. In football specifically, machine learning and reinforcement learning have been used for tactical decision analysis, activity recognition, player-motion recognition, and set-piece recommendation [15,2,23,24]. These studies demonstrate the growing importance of models capable of learning complex, contextual dependencies from football

data. However, their practical adoption depends not only on predictive quality but also on the transparency of the resulting decisions.

Transformer models offer an attractive architecture for modeling sequential football events. Based on self-attention, they allow each event in a sequence to interact with other events and can capture long-range dependencies without processing the sequence strictly step by step [21]. This is particularly useful in football, where an early pass, carry, or positional action may influence the quality of a later shooting opportunity. In addition, attention weights can be extracted and analyzed, making Transformer-based models potentially useful for interpretable sports analytics.

### 2.3 Attention and Interpretability

Attention mechanisms were originally introduced to improve the performance of neural models by allowing them to focus on relevant parts of the input [21]. In many applications, attention weights have also been used as explanations. The basic intuition is simple: if a model assigns a high attention weight to an input element, that element may be important for the model’s prediction.

However, attention-based explanations have important limitations. Attention weights do not necessarily imply causality, and the relationship between attention and explanation remains debated [7,25]. A model may attend to an input feature for reasons that are not aligned with human interpretation. Moreover, different models can produce similar predictions with different attention distributions. For this reason, attention should be validated before being used as an explanatory tool. In football analytics, this issue is especially important because analysts and coaches may use model explanations to support decisions about players, tactics, or scouting.

The present work builds on action-valuation and sequence-modeling research in football, but shifts the focus from assigning value to isolated actions to deriving interpretable player contribution scores from attention distributions over complete shot-ending sequences. This positioning also distinguishes our approach from standard xG, xT, VAEP, and OBV models: instead of replacing these metrics, the proposed framework uses the supervised prediction of shot quality as a learning signal and then analyzes the attention structure that emerges from the sequence model.

## 3 Methodology

### 3.1 Dataset and Shot Sequence Construction

The empirical analysis uses event data from the Polish Ekstraklasa 2025/2026 season, with event structures and coordinate conventions consistent with StatsBomb data [20]. The analysis was conducted on the full available dataset for the season, without filtering matches by the participation of any particular team.

After filtering the competition and season, 306 matches were identified and included in the study. For these matches, shot-ending possessions were extracted

and transformed into event sequences. The final sample contains 8,441 shot sequences involving actions of 517 players. Although the dataset remains moderate in size for a neural sequence model, it is sufficiently diverse to support a league-wide proof-of-concept evaluation of the proposed attention-based framework. We therefore interpret the results as evidence of the method’s feasibility and analytical usefulness under realistic event-data conditions, while avoiding claims of generalization beyond the analyzed competition and season.

For each shot, we construct the possession sequence of the shooting team up to and including the shot. Each sequence contains at most 20 events; shorter sequences are padded, and longer sequences are truncated to retain the most recent events. Each event token may contain the player identity, event type, event outcome, start location, end location, temporal difference relative to the previous event, and positional order. Spatial coordinates are normalized to the pitch dimensions  $120 \times 80$ , and temporal differences are transformed using a logarithmic scaling.

### 3.2 Target–Feature Relationship and Leakage-Aware Framing

The supervised target is the provider xG value of the final shot. This design is useful for learning a sequence representation tied to chance quality, but it also creates an important methodological caveat. Since the sequence includes the final shot token, and the shot token may include shot-specific information such as shot type, location, end location, or outcome, the task should not be interpreted as independent pre-shot forecasting. In particular, a model that observes the final shot token may partially reconstruct the source xG model rather than infer shot quality only from the preceding buildup.

For this reason, we explicitly frame the present experiment as a *post-hoc attention audit* of an observed shot-ending sequence. The target is used as a representation-learning signal, and the attention distribution is analyzed as a model-reliance signal. We do not claim that the attention weights are causal effects, nor that the model provides a validated alternative to xG, xT, VAEP, or OBV. A leakage-free predictive variant would require excluding the shot token or masking its shot-specific features. We regard such variants as necessary for future confirmatory evaluation.

Formally, let a shot-ending sequence be represented as

$$S = (e_1, e_2, \dots, e_T), \quad (1)$$

where  $e_T$  is the final shot event and  $e_1, \dots, e_{T-1}$  are the preceding events in the same possession. The model estimates

$$\hat{y} = f(S), \quad (2)$$

where  $y$  denotes the observed xG value of the final shot and  $\hat{y}$  denotes its model-based prediction. In the present setting, this prediction task is best understood as a proxy task for learning an attention-bearing representation of the complete observed sequence.

### 3.3 Temporal Shot Transformer

The proposed model is a Transformer-based sequence model [21]. Categorical features are encoded through embeddings for player identity, event type, and event outcome. Continuous features, including spatial coordinates and temporal differences, are processed by a feed-forward layer. Positional embeddings are added to retain order information.

The token representations are passed through a Transformer encoder. The final token representation is used by a regression head to estimate the xG target. In the experiments reported here, the model used an embedding dimension of 256, eight attention heads, four encoder layers, a dropout equal to 0.1, the AdamW optimizer, and a batch size of 64. Training was performed with early stopping rather than a fixed small number of epochs. The maximum number of epochs was set to 100, with patience of 10 epochs and a minimum improvement threshold of  $10^{-5}$  in validation MSE. The validation split was defined at the match level, so that sequences from the same match were not shared between the training and validation sets. After training, the model parameters from the epoch with the lowest validation MSE were restored and used for all subsequent attention-based analyses. The final model contained 3,265,265 trainable parameters. Most parameters were located in the four Transformer encoder blocks, especially in the self-attention projections and feed-forward sublayers, while the player, event-type, outcome, continuous-feature, positional, and regression-head components accounted for the remaining parameters.

This architecture is relatively large for the available data. We keep it in order to analyze multi-head self-attention, but we explicitly discuss overfitting, validation-set instability, and statistical fragility as threats to validity.

### 3.4 Attention Extraction and Player Contribution

Attention maps are extracted from the trained Transformer. We focus on the attention distribution associated with the final shot token. Attention maps are averaged over the last two Transformer layers and over attention heads. Padding tokens are excluded, and the remaining weights are renormalized. Let  $a_t$  denote the attention assigned to event  $e_t$ :

$$\sum_{t=1}^T a_t = 1. \quad (3)$$

The attention weights are interpreted conservatively as model-reliance scores. To estimate player contribution, token-level attention weights are aggregated by player. If player  $p$  is involved in events indexed by  $I_p$ , the contribution of player  $p$  in a sequence is

$$C_p(S) = \sum_{t \in I_p} a_t. \quad (4)$$

To summarize the contribution across the analyzed sample, let  $S_i$  denote the  $i$ -th shot-ending sequence and let  $y_i$  denote the observed xG value of its

final shot. Let  $\mathcal{D}_p$  be the set of sequences in which player  $p$  appears. The total contribution of player  $p$  is then defined as

$$C_p^{\text{total}} = \sum_{S_i \in \mathcal{D}_p} C_p(S_i), \quad (5)$$

whereas the xG-weighted contribution is defined as

$$C_p^{\text{xG}} = \sum_{S_i \in \mathcal{D}_p} y_i C_p(S_i). \quad (6)$$

The xG-weighted score emphasizes player involvement in sequences that ended with higher-quality shooting opportunities.

The aggregation may include the final shot event if the shooter appears as a valid player token. Therefore, the resulting score should be read as a contribution to the complete shot-ending sequence, not purely as a buildup contribution before the shot.

We report several player-level summaries: number of shot sequences involving the player, total contribution, average contribution per shot sequence, total xG of sequences involving the player, and xG-weighted contribution. To reduce sample-size artifacts, rankings based on average contribution should be interpreted together with the number of sequences in which the player appears.

### 3.5 Experimental Setup and Baselines

The 8,441 shot-ending sequences are split into 7,175 training and 1,266 validation examples. Results are averaged over five random seeds. Predictive performance is measured using mean squared error. Interpretability-oriented diagnostics include mean attention entropy, Spearman rank stability relative to the baseline ranking, and Jaccard@10 similarity of the top-player sets.

The main comparison is an ablation study:

- *baseline*: full model with all available input components;
- *no\_player*: player identity removed;
- *no\_spatial*: spatial features removed;
- *no\_temporal*: temporal-difference feature removed;
- *no\_outcome*: event outcome removed;
- *uniform control*: learned attention replaced by uniform attention for contribution aggregation.

The uniform control and the number of shot sequences involving each player serve as simple descriptive baselines for interpreting player rankings. We acknowledge that stronger predictive and attribution baselines—for example, gradient boosting on aggregated sequence features, recurrent sequence models, occlusion-based contribution, integrated gradients, SHAP, or action-value metrics such as VAEP/xT/OBV—are needed for a full competitive evaluation. The present study, therefore, emphasizes transparency and diagnostic analysis rather than superiority over existing valuation frameworks.

## 4 Results

### 4.1 Predictive Performance and Attention Entropy

Table 1 presents the main ablation results. The baseline model achieves a validation mean squared error of  $0.0123 \pm 0.0008$ . Removing outcome-related information results in the greatest degradation, increasing the validation mean squared error to  $0.0175 \pm 0.0033$ . This indicates that event outcomes carry a strong predictive signal in the current representation.

At the same time, removing player identity, spatial context, or temporal information does not clearly harm the validation error. The *no\_player* variant obtains  $0.0116 \pm 0.0014$ , while *no\_spatial* and *no\_temporal* obtain  $0.0121 \pm 0.0010$  and  $0.0119 \pm 0.0007$ , respectively. These values should not be interpreted as evidence that such components are irrelevant. Given the small validation set and the potential availability of strong target-related information in the final shot token, these results may also reflect redundancy, overfitting, or leakage from other features. For this reason, we treat the ablation table as diagnostic rather than confirmatory.

**Table 1.** Ablation results for predictive performance and attention entropy. Values are reported as mean  $\pm$  standard deviation over five random seeds. Bold indicates the largest degradation in validation error relative to the baseline. The uniform control uses the baseline predictions and replaces learned attention weights only for contribution aggregation.

| Variant         | Train MSE           | Validation MSE                        | Attention entropy   |
|-----------------|---------------------|---------------------------------------|---------------------|
| baseline        | $0.0098 \pm 0.0003$ | $0.0123 \pm 0.0008$                   | $2.1654 \pm 0.0014$ |
| no_outcome      | $0.0142 \pm 0.0007$ | <b><math>0.0175 \pm 0.0033</math></b> | $2.1612 \pm 0.0023$ |
| no_player       | $0.0116 \pm 0.0004$ | $0.0116 \pm 0.0014$                   | $2.1664 \pm 0.0026$ |
| no_spatial      | $0.0098 \pm 0.0006$ | $0.0121 \pm 0.0010$                   | $2.1642 \pm 0.0033$ |
| no_temporal     | $0.0097 \pm 0.0004$ | $0.0119 \pm 0.0007$                   | $2.1651 \pm 0.0012$ |
| uniform control | same as baseline    | same as baseline                      | $2.1710 \pm 0.0000$ |

The attention entropy values are relatively similar across the baseline and most ablation variants. The baseline entropy is  $2.1654 \pm 0.0014$ . The uniform attention control has the highest entropy, 2.1710, as expected, since a uniform distribution is maximally dispersed. However, the small differences among the learned attention variants indicate that entropy alone is insufficient to evaluate interpretability. A model may maintain a similar level of attentional concentration while assigning different levels of importance to different players. Therefore, ranking stability provides a more informative evaluation.

## 4.2 Stability of Attention-Based Rankings

Table 2 presents the stability of normalized player-contribution rankings relative to the learned-attention baseline. The rankings are based on mean contribution rather than total accumulated contribution, because total scores are strongly affected by the frequency with which a player appears in shot-ending sequences.

The *no\_player* variant yields the largest drop in stability among model ablations. Its Jaccard@10 value is  $0.8545 \pm 0.0727$ , indicating that removing player identity changes the set of players identified as highly influential on a per-action basis. In contrast, *no\_outcome* has a higher Jaccard@10 of  $0.9636 \pm 0.0727$ , while *no\_spatial* and *no\_temporal* both obtain  $0.9273 \pm 0.0891$ .

The uniform control obtains a Jaccard@10 of 0.8545, showing that replacing learned attention with equal token weights can change the top-player set even when the global rank correlation remains high. This supports the use of both Spearman correlation and Jaccard@10: the former captures overall ranking similarity, whereas the latter is more sensitive to changes among the top-ranked players.

**Table 2.** Stability of normalized player-contribution rankings relative to the learned-attention baseline. Spearman correlations and Jaccard@10 similarities are reported as mean  $\pm$  standard deviation over five random seeds. Bold indicates the largest drop in stability among model ablations. The uniform control uses the same sequences but replaces learned attention weights with uniform weights when aggregating contributions.

| Variant         | Spearman mean                         | Jaccard@10 mean                       |
|-----------------|---------------------------------------|---------------------------------------|
| no_outcome      | $0.9968 \pm 0.0004$                   | $0.9636 \pm 0.0727$                   |
| no_player       | <b><math>0.9953 \pm 0.0019</math></b> | <b><math>0.8545 \pm 0.0727</math></b> |
| no_spatial      | $0.9982 \pm 0.0007$                   | $0.9273 \pm 0.0891$                   |
| no_temporal     | $0.9994 \pm 0.0002$                   | $0.9273 \pm 0.0891$                   |
| uniform control | $0.9970 \pm 0.0000$                   | $0.8545 \pm 0.0000$                   |

These findings suggest that the components most important for predictive performance are not necessarily the same as those most important for stable player-level explanations. Outcome information is most important for validation error, whereas player identity has the strongest effect on the stability of normalized player rankings.

## 4.3 Player Contribution Rankings

The player report provides several complementary rankings. When players are ranked by total contribution, the top players are those who appear frequently in shot-ending sequences and accumulate high total attention. Table 3 shows the ten highest-ranked players by xG-weighted attention contribution, while the Total column reports their accumulated unweighted attention contribution across

all shot-ending sequences. The results align well with independent end-of-season assessments. Tomáš Bobček, the highest-ranked player by xG-weighted attention contribution, was the 2025/2026 Ekstraklasa top scorer, while Bartosz Nowak, who had the highest total attention contribution and ranked fourth by the xG-weighted metric, was named Player of the Season and finished as the league assist leader. This supports the interpretation that the proposed attention-based measure identifies players with substantial involvement in high-value attacking sequences, rather than merely reflecting raw participation volume. The ranking also includes players from lower-ranked clubs, suggesting that the metric captures individual offensive importance within a team context. The goals-plus-assists values shown next to player names provide a simple external reference for the attention-based ranking.

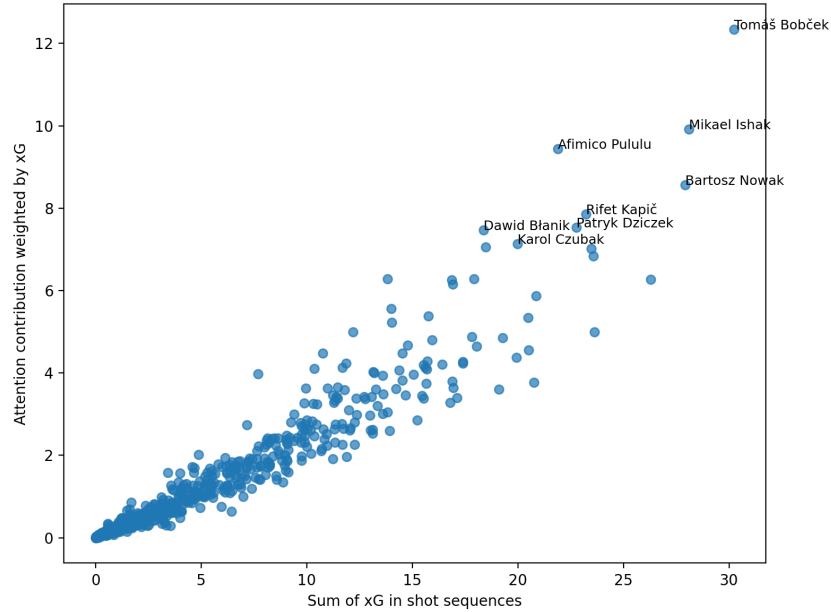
**Table 3.** Top 10 players by xG-weighted attention contribution. Values in parentheses after player names denote league goals plus assists in the 2025/2026 Ekstraklasa season. Club names refer to the clubs represented in the analyzed match sample, and values in parentheses after club names denote the club’s final league position.

| Player (G+A)           | Club (final position)     | Sequences  | Total          | Mean xG-wtd.               |
|------------------------|---------------------------|------------|----------------|----------------------------|
| Tomáš Bobček (20+6)    | Lechia Gdańsk (16)        | 201        | 69.785         | <b>0.347</b> <b>12.342</b> |
| Mikael Ishak (16+5)    | Lech Poznań (1)           | 201        | 52.987         | 0.264 9.918                |
| Afimico Pululu (15+5)  | Jagiellonia Białystok (3) | 167        | 50.460         | 0.302 9.440                |
| Bartosz Nowak (9+12)   | GKS Katowice (5)          | <b>300</b> | <b>100.325</b> | 0.334 8.567                |
| Rifet Kapić (5+3)      | Lechia Gdańsk (16)        | 227        | 66.861         | 0.295 7.852                |
| Patryk Dzik (4+3)      | Piast Gliwice (15)        | 259        | 68.772         | 0.266 7.535                |
| Dawid Błanik (9+4)     | Korona Kielce (11)        | 188        | 64.568         | 0.343 7.465                |
| Karol Czubak (18+3)    | Motor Lublin (12)         | 154        | 43.880         | 0.285 7.128                |
| J. Braut-Brunes (16+0) | Raków Częstochowa (4)     | 148        | 39.283         | 0.265 7.056                |
| Kamil Grosicki (6+5)   | Pogoń Szczecin (9)        | 231        | 68.466         | 0.296 7.022                |

This ranking should be interpreted as a measure of global involvement in influential attacking sequences. It rewards both repeated participation and high attention weights. Players with many appearances in shot-ending sequences naturally have more opportunities to accumulate contributions. We therefore avoid using average contribution alone as the primary criterion for player ranking. In small samples, average-per-shot rankings can be dominated by players with very few appearances. A practical deployment should report minimum-involvement thresholds and confidence intervals or bootstrap uncertainty bands for all player-level summaries.

A different picture emerges when players are ranked by average contribution per shot. In this case, some players with very few sequences appear at the top. This illustrates the importance of considering sample size when interpreting per-action metrics. The expected-goals-weighted contribution ranking provides another perspective. It emphasizes players involved in sequences that lead to high-quality chances. This metric may be particularly useful for identify-

ing players whose involvement is associated with dangerous attacking situations. Figure 1 illustrates the relationship between the total expected goals of sequences involving a player and the expected-goals-weighted contribution.



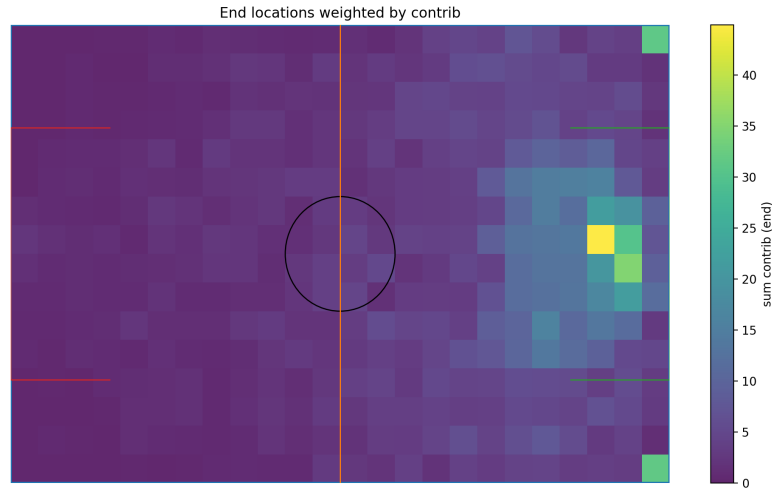
**Fig. 1.** Relationship between the total expected goals of shot sequences involving a player and the expected-goals-weighted contribution. Labeled points correspond to players with the largest combined attacking impact.

#### 4.4 Visual Interpretation

The framework also generates visual outputs to support qualitative analysis. Attention-weighted heatmaps can show where influential events occur on the pitch. Goal action plots can show the structure of selected scoring actions and highlight which events received high attention. In the final manuscript, we focus on two visual forms that are both interpretable and legible in print: a spatial heatmap of end locations weighted by contribution and a single action-level explanation of a goal sequence.

Figure 2 shows end locations weighted by attention-based contribution after standardizing all attacking sequences to a left-to-right direction. Figure 3 illustrates how attention can be inspected at the level of a single shot-ending sequence. The highest contribution to the pre-shot sequence (28% of the total attention) was assigned by the model to Filip Jagiełło, a midfielder for Lech

Poznań. From an analytical and coaching perspective, this is well-justified by the structure of the attacking move: after receiving the ball from G. Þórðarson, F. Jagiełło executes a crucial, long progressive carry (indicated by the dashed orange line), dynamically gaining territory in the central zone of the pitch. This sequence is crowned with a precise progressive pass (indicated by the thick blue line) into the high-threat under-shot zone, directly setting up the final strike converted into a goal by Leo Bengtsson. Involvement in the final phase of the build-up and shot preparation is also reflected in the high attention score for M. Ishak (24%). This case demonstrates that the model can successfully identify and value progressive actions that actively destabilize the opponent’s defensive structure, shifting the analytical focus from just the final shooters and assist-makers to the key space creators in the transition phase.

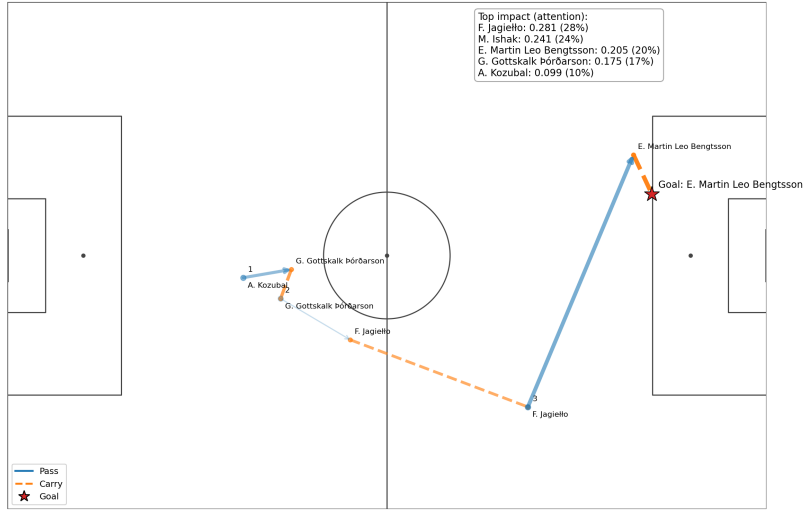


**Fig. 2.** End locations weighted by attention-based contribution. The heatmap summarizes where influential actions tend to terminate in the analyzed shot-ending sequences.

Such visualizations are important because player rankings alone may not fully explain how contributions arise. A player may be highly ranked because of repeated involvement in buildup, because of decisive actions near the penalty area, or because of participation in high-expected-goals sequences. Visual inspection allows analysts to distinguish between these cases.

## 5 Discussion and Conclusion

The empirical results support a modest conclusion: attention-derived contribution scores can be constructed for football event sequences and inspected via rankings, stability diagnostics, and visualizations. They do not yet support a



**Fig. 3.** Example of an attention-based explanation for a single goal action. Line width represents attention intensity assigned to events in the sequence leading to the goal.

strong claim that the proposed model is a validated player-rating system or that attention weights provide objectively correct explanations.

The clearest quantitative pattern concerns the difference between predictive and interpretability-oriented diagnostics. Removing outcome-related information produces the largest degradation in validation mean squared error, whereas removing player identity has a stronger effect on the stability of mean-per-token top-player rankings. This suggests that the information components that help a model fit the xG proxy task are not necessarily the same as those that stabilize player-level explanations. This distinction is useful for interpretable AI in sport because it clarifies why evaluation should encompass more than prediction error alone.

The study also highlights several threats to validity. First, the target is a source-provided xG value, and the final shot token is part of the input sequence. As a result, the model may partially learn to reconstruct the provider xG model rather than infer chance quality from the preceding buildup. This is not a minor technicality: it directly affects how attention should be interpreted. The present results should therefore be read as a post-hoc attention audit of observed shot-ending sequences. A stronger future design should compare at least three settings: preceding events only, preceding events plus masked shot-location information, and the full observed sequence.

Second, the dataset contains 8,441 shot-ending sequences, which remains moderate for a Transformer with 256-dimensional embeddings, eight attention heads, and four encoder layers. Using five random seeds reduces sensitivity to

initialization but does not fully eliminate statistical fragility, especially when deriving player-level rankings from attention-based attribution scores. Future work should further strengthen the empirical validation by incorporating multiple seasons, larger cross-season samples, and temporal validation splits that better reflect prospective deployment conditions.

Third, the current evaluation lacks strong external baselines and independent validation of explanations. Uniform attention and involvement counts are useful sanity checks, but a competitive evaluation should include simpler predictive models, recurrent sequence models, gradient-boosted aggregation baselines, action-value baselines inspired by VAEP/xT/OBV, and perturbation-based attribution methods. It should also compare attention-derived rankings with expert analyst judgment or established football metrics.

Fourth, the player rankings are sample-dependent. Total contribution is affected by frequency of involvement, while average contribution can overemphasize players with few sequences. For practical use, contribution scores should be reported with minimum-involvement thresholds and uncertainty estimates.

Despite these limitations, the framework is useful as an interpretable analysis pipeline. It shows how event sequences can be converted into player-level contribution scores, how those scores can be stress-tested under ablations, and how attention can be visualized at aggregate and single-action levels. The main methodological message is that attention in sports analytics should not be accepted as explanation by default; it should be treated as a hypothesis-generating signal that requires leakage checks, baseline comparisons, stability analysis, and domain validation.

In conclusion, this paper presents a transparent case study of attention-based player contribution in football event sequences. The proposed method links a sequence model, attention extraction, player-level aggregation, and visual analytics. The results are promising but preliminary. Future work will focus on leakage-free pre-shot modeling, larger temporal evaluations, stronger baselines, and validation by football analysts.

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